



USN

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (ISE) (Credit System) Degree Examinations

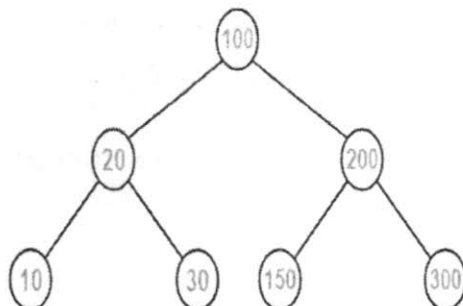
Supplementary Examinations – December 2025

21CS302 / 20CS305 / 21IS302 / 20IS305 / 21AM302 / 20AM305 / 21CC302 / 20CC305 – DATA STRUCTURES
 Duration: 3 Hours Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1.	a) Write a C program to implement stack using arrays.	12	L*3	1	2
	b) What are the applications of stack?	4	L2	1	1
	c) Write a C Program to compute sum of the array elements using pointers.	4	L3	1	2
2.	a) Write a C program to evaluate a postfix expression.	8	L3	1	2
	b) How data structures are classified?	6	L2	1	1
	c) Write a C program to swap two numbers using call by reference.	6	L3	1	2
3.	a) What is recursion? Write a program to find Fibonacci of a number using recursion.	8	L3	1	2
	b) Obtain the postfix expression for ((A+(B-C) *D)^E+F) by substitution method:	6	L3	1	2
	c) Write a C program to demonstrate decrementing a pointer.	6	L3	1	2
Unit – II					
4.	a) What are the advantages of circular queue? Write a function to perform the following operations on circular queue. i) Insert ii) Delete iii) Display	10	L2	2	1
	b) How do you represent Doubly linked list in C. Write functions to i) Insert a node at the front end ii) Delete a node from the rear end.	10	L2	2	1
5.	a) Write a function to perform following operations on singly linked list i) insert rear ii) delete front iii) Display	10	L2	2	1
	b) Write a C program to implement ordinary queue.	10	L3	2	2
6.	a) Write a function to perform following operations on circular singly linked list with header node i) insert front ii) insert rear iii) Display	12	L2	2	1
	b) What is priority queue? Explain the types of priority queue.	8	L2	2	1
Unit – III					
7.	a) What is threaded binary tree? Explain the types of threaded binary tree.	8	L2	3	1

- b) What is preorder, inorder and postorder traversal of the following tree?



- c) Write a function to insert an item into binary search tree.
8. a) Explain the following graph terminologies with example
 i) Acyclic graph ii) Connected graph iii) Complete graph
 iv) Multigraph
- b) Construct an expression tree for the following expression
 $a + (b * c) + d * (e + f)$
- c) Discuss different types of Hashing.

6	L3	3	2
6	L2	3	1

8	L2	3	1
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6	L3	3	2
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6	L2	3	1
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BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
